

# Deepak Mallampati

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Kent, OH

## PROFESSIONAL SUMMARY

Applied AI engineer with shipped computer vision + video understanding + production deep-learning discipline - YOLOv8 real-time driver-drowsiness detection-and-classification pipeline (replaced two-stage CNN with ~30% inference latency reduction and ~25% false-positive reduction) and transformer-based hierarchical video summarization across 5,000+ recorded sessions (60-70% review-time reduction, ~85% alignment with human-curated highlights). MSc Computer Science (Kent State). Strong Python + FastAPI/Docker production discipline with structured logging, audit trails, and ~30% post-deploy defect reduction at Progress Solutions. US-based, F-1 OPT - open to discussing EU work authorization.

## CORE COMPETENCIES

Applied AI / LLMs • RAG + Grounding • Agentic Workflows • Schema Contracts • Eval Harnesses • Python / FastAPI • TypeScript / React • Docker / CI-CD • Multimodal Vision • Predictive ML

## PROFESSIONAL EXPERIENCE

### Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Agent Engineering, Production LLM

USA - Healthcare Technology, AI Consulting

- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval; recursive semantic chunking and transformer embeddings improved retrieval precision ~**35%**, reduced irrelevant retrieval >**30%**, response alignment >**90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries with tool discipline, structured reasoning, and grounding rules; response stability ~**25%**, hallucinations cut >**30%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show and care engagement scoring; recall +**15-20%** on high-risk categories, precision held >**90%** via class weighting, stratified sampling, threshold calibration.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation; post-deploy defects ~**30%**.
- EHR preprocessing (Pandas, NumPy); dataset reliability >**98%**, downstream instability **-40%**.
- HIPAA-conscious governance: de-identification, data lineage, evaluation audit trails, stakeholder-facing system-limitation docs.

### Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas ERP

- Optimized **T-SQL stored procedures** on a compliance-sensitive ERP; query time **-20%**, retrieval latency **-25%**.
- Built **Jenkins CI/CD pipelines** for schema updates and stored-procedure deployments; deploy errors **-30%**, release cycle **-35 to -40%**.
- Designed performance dashboards with **SQL Server DMVs + Grafana**; incident recurrence **-25%**, root-cause resolution **-30%**.
- Tuned index maintenance and partition-aware queries; reconciliation runtime **-15%**, deadlocks **-15%**.

### Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India - Nonprofit, Video Analytics, Applied NLP

- Built **transformer-based video summarization** over 5,000+ recorded sessions; hierarchical summarization and timestamp-aligned clip extraction cut manual review **60-70%**.
- AI-selected highlights aligned with human-curated moments ~**85%**.
- Implemented **document Q&A with RAG**; hallucinations ~**30%**, contextual accuracy >**85%**.
- Deployed as **Flask** API with a lightweight web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

5-stage multi-agent pipeline (Intake, Validation, Consistency, Duplicate, Risk Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policies. Audit-ready explainable risk scoring.  
*Python, LangChain, FastAPI, pgvector, GPT-4, schema contracts*

Manga Lens - Multi-Provider AI Vision Chrome Extension

Real-time manga translation extension shipped to the Chrome Web Store. Multi-provider abstraction (4 AI vision providers) with per-provider payload handling for worst-case failure isolation. 29 sites supported across Manifest V3 service workers.  
*TypeScript, Manifest V3, Service Workers, OpenAI / Anthropic / Google / OpenRouter*

Dream Decoder - Multimodal Creative Platform

End-to-end multimodal pipeline: text description, intermediate prompt transformation, grounded image generation. ~30% contextual alignment gain, ~25-30% first-pass image success.  
*FastAPI, React + Vite, Tailwind, Perplexity Sonar, Replicate*

YOLOv8 Driver Drowsiness Detection

Real-time computer vision for driver drowsiness; sub-50ms inference on edge hardware. Walk-forward validation and class-weighted training for imbalanced safety data.  
*PyTorch, YOLOv8, OpenCV, ONNX*

EDUCATION

|                                                                                       |                     |
|---------------------------------------------------------------------------------------|---------------------|
| <b>Master of Science, Computer Science</b> - Kent State University                    | Aug 2023 - May 2025 |
| Focus: Applied Machine Learning, NLP, Database Systems.                               |                     |
| <b>Bachelor of Technology, Computer Science</b> - KL University                       | Jun 2019 - May 2023 |
| Founder / Lead of E-Farming platform (university project, pilot with 3 cooperatives). |                     |

CERTIFICATIONS

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|-----------------------------------------------------------|------|
| Deep Learning Specialization - DeepLearning.AI (Coursera) | 2024 |
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SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), Java, C++  
**AI / ML:** PyTorch, scikit-learn, XGBoost, LangChain, LlamaIndex, OpenAI, Anthropic, Hugging Face Transformers, Diffusers, YOLOv8  
**LLM / GenAI:** RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, embeddings, vector search  
**Backend:** FastAPI, Flask, REST, WebSockets, Docker, Celery, Playwright  
**Frontend:** React, Next.js, Vite, Tailwind, Chrome Extensions (Manifest V3)  
**Data:** PostgreSQL + pgvector, MongoDB, Redis, Pandas, NumPy, Snowflake  
**DevOps:** Docker, Jenkins, GitHub Actions, Grafana, Prometheus, AWS (EC2, S3, Lambda)  
**Domains:** Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal